RS-WS-N01-BYH-EX User manual of louver box temperature and humidity transmitter (Type 485)

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Table of Contents

1. product description 4
2 Equipment installation instructions 5
3. Configuration software installation and use 7
4. letter of agreement 8
5. Common problems and solutions 10
6. Contact information 11
7. Document history 11
Appendix: Shell size 11

1. product description

1.1product description

The transmitter is widely used in communication rooms, warehouse buildings and automatic control and other places that require temperature and humidity monitoring. It is safe and reliable, beautiful in appearance and easy to install.

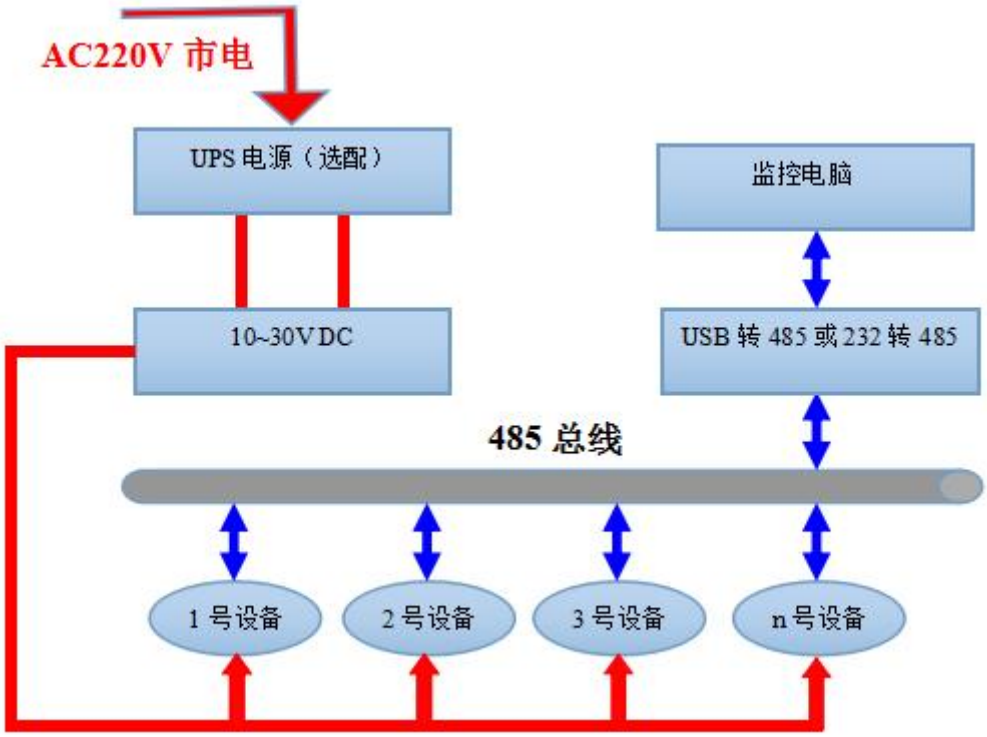
1.2Features

The measuring unit imported from Switzerland is adopted for accurate measurement. Using a dedicated 485 circuit, the communication is stable. 10~30V wide voltage range power supply, complete specifications and easy installation.

1.3The main technical parameters

DC power supply (default)	DC 10-30V	
Maximum power consumption	0.4W	
A precision	humidity	±2%RH(60%RH,25℃)
	temperature	±0.4℃（25℃）
B precision(default)	humidity	±3%RH(60%RH,25℃)
	temperature	±0.5℃（25℃）
Transmitter circuit operating temperature	-20℃~+60℃，0%RH~80%RH	
Measurement range	-40℃~+80℃，0%RH~100%RH	
Temperature display resolution	0.1℃	
Humidity resolution	0.1%RH	
Fresh time	1s	
Long-term stability	humidity	≤1%RH/y
	temperature	≤0.1℃/y
Response time	humidity	≤8s(1m/s Wind speed)
	temperature	≤25s(1m/s Wind speed)
Signal output type	RS485(Modbus protocol)	
	Bracket installation	

1.4system framework



System scheme block diagram

2. Equipment installation instructions

2.1Inspection before equipment installation

Equipment List:

1 set of temperature and humidity transmitter equipment

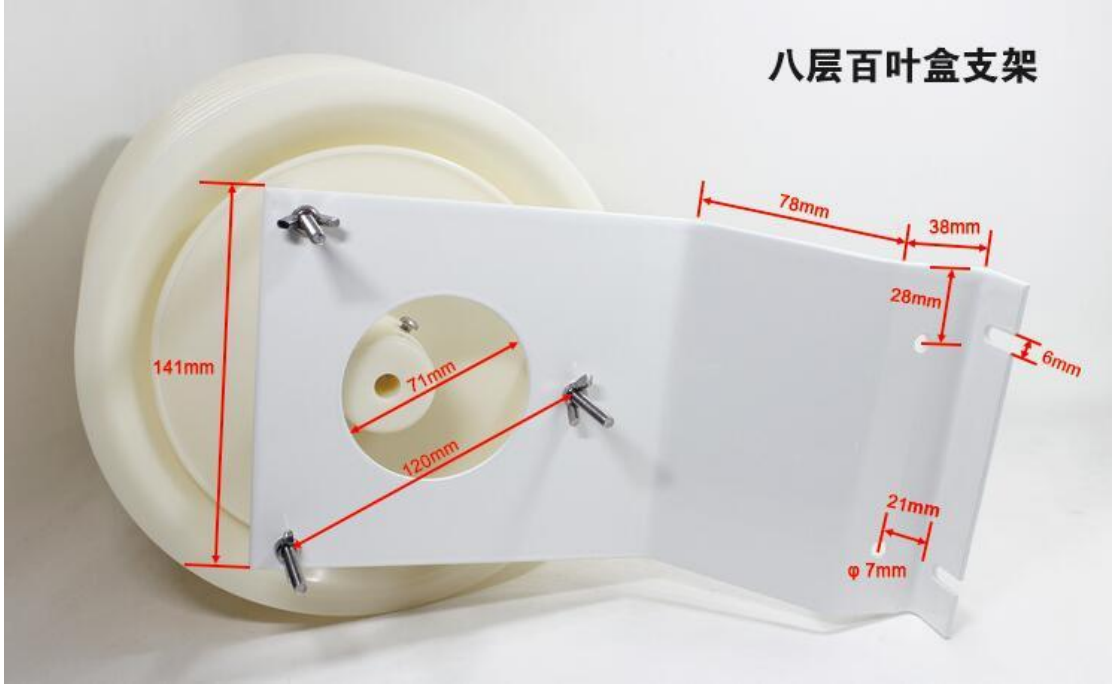
Certificate of conformity, warranty card, calibration report, etc.

2 mounting screws

USB to 485 (optional)

485 terminal resistance (gift for multiple devices)

2.2 Installation size



Special Note:

1）There are certain specification requirements for 485 field wiring. For details, please refer to the data package "485 Equipment Field Wiring Manual".

2）When the device is connected to the 485 bus, ensure that the addresses of multiple devices will not be repeated.

2.3Wiring instructions

Power and 485 signal

Wide-voltage power input can be 10~30V. When wiring the 485 signal line, pay attention to the two wires A and B that cannot be reversed, and the addresses of multiple devices on the bus cannot conflict.

2.4 Specific wiring

	Thread color	Description
power supply	brown	Positive power supply (10~30V DC)
	black	Power negative
Communication	yellow	485-A
	blue	485-B

3. Configuration software installation and use

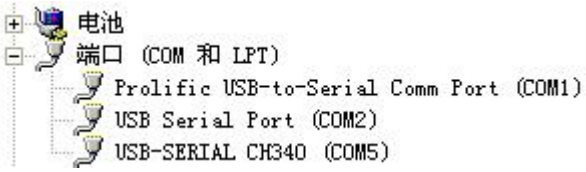
3.1 Software selection

Open the data package, select "Debugging software" --- "485 parameter configuration software", find

Just open it. Note: When using the configuration software to change the address and baud rate, only one device can be connected.

3.2 parameter settings

① Select the correct COM port (check the COM port in "My Computer—Properties—Device Manager—Port"). The following figure lists the driver names of several different 485 converters.



② Connect only one device alone and power it on, click the test baud rate of the software, the software will test the baud rate and address of the current device, the default baud rate is 4800bit/s, and the default address is 0x01.

③. Modify the address and baud rate according to the needs of use, and at the same time query the current function status of the device.

④. If the test is unsuccessful, please recheck the equipment wiring and 485 driver installation.

4. letter of agreement

4.1 Basic communication parameters

Code	8-bit binary
Data bit	8-bit
Parity bit	no
Stop bit	1 person
Error checking	CRC (Redundant Cyclic Code)
Baud rate	2400bit/s, 4800bit/s, 9600 bit/s can be set, the factory default is 4800bit/s

4.2Data frame format definition

Using Modbus-RTU communication protocol, the format is as follows:

Initial structure ≥ 4 bytes time

Address code = 1 byte

Function code = 1 byte

Data area = N bytes

Error check = 16-bit CRC code

Time to end structure ≥ 4 bytes

Address code: the address of the transmitter, which is unique in the communication network (factory default 0x01).

Function code: the command function instruction issued by the host, this transmitter only uses function code 0x03 (read register data).

Data area: The data area is the specific communication data, pay attention to the high byte of 16bits data first!

CRC code: two-byte check code.

Host query frame structure:

address code	function code	Register start address	Register length	Check code low bit	Check code high
1byte	1byte	2byte	2byte	1byte	1byte

Slave machine response frame structure:

address code	function code	Effective bytes	Data area	Second data area	Nth data area	Check code
1byte	1byte	1byte	2byte	2byte	2byte	2byte

4.3 Register address

Register address	PLC or configuration address	content	operating
0000 H	40001	humidity	Read only
0001 H	40002	temperature	Read only

4.4 Communication protocol example and explanation

Example: Read the temperature and humidity value of the device address 0x01

Inquiry frame (hexadecimal):

address code	function code	initial address	Data length	Check code low bit	Check code high
0x01	0x03	0x00 0x00	0x00 0x02	0xC4	0x0B

Response frame (hexadecimal): (for example, the temperature is -10.1℃ and the humidity is 65.8%RH)

address code	function code	Returns the number of valid bytes	Humidity value	Temperature value	Check code low bit	Check code high
0x01	0x03	0x04	0x02 0x92	0xFF 0x9B	0x5A	0x3D

Temperature calculation:

When the temperature is lower than 0 ℃, the temperature data is uploaded in the form of complement code.

Temperature: FF9B H (hexadecimal) = -101 => temperature = -10.1℃

Humidity calculation:

Humidity: 292 H (hexadecimal) = 658 => Humidity = 65.8%RH

5. Common problems and solutions

The device cannot connect to the PLC or computer

possible reason:

1) The computer has multiple COM ports, the selected port is incorrect

2) The device address is wrong, or there are devices with duplicate addresses (the factory default is all 1).

3) The baud rate, check method, data bit, stop bit are wrong.

4) The 485 bus is disconnected, or the A and B wires are reversed

5) If the number of equipment is too large or the wiring is too long, power supply should be nearby, add 485 booster, and add 120Ω terminal resistance.

6) USB to 485 driver is not installed or damaged

7) The equipment is damaged.

6. Contact information

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7. Document history

V1.0 document creation
V2.0 documentation update